11/2/2026: Intro to Probability (Part 1)

CSCI 246: Discrete Structures

Textbook reference: Sec 30, Scheinerman

Graded Quiz Pickup

Quizzes are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Today's Agenda

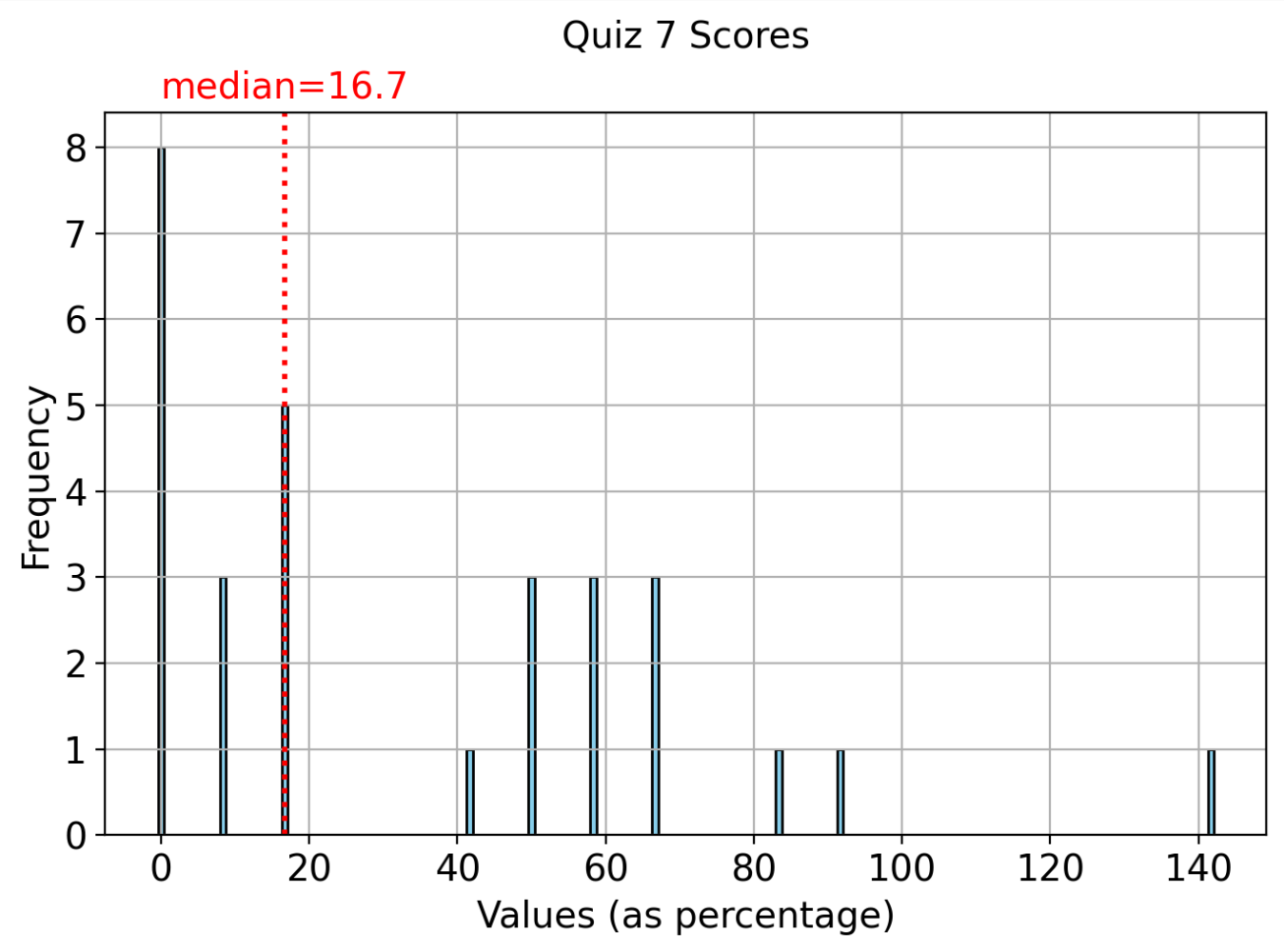
- Follow up from last class ($\approx$ 10 mins)
- Mini-lecture ($\approx$ 25 mins)
- Group exercises ($\approx$ 15 mins)

Follow-up from last class

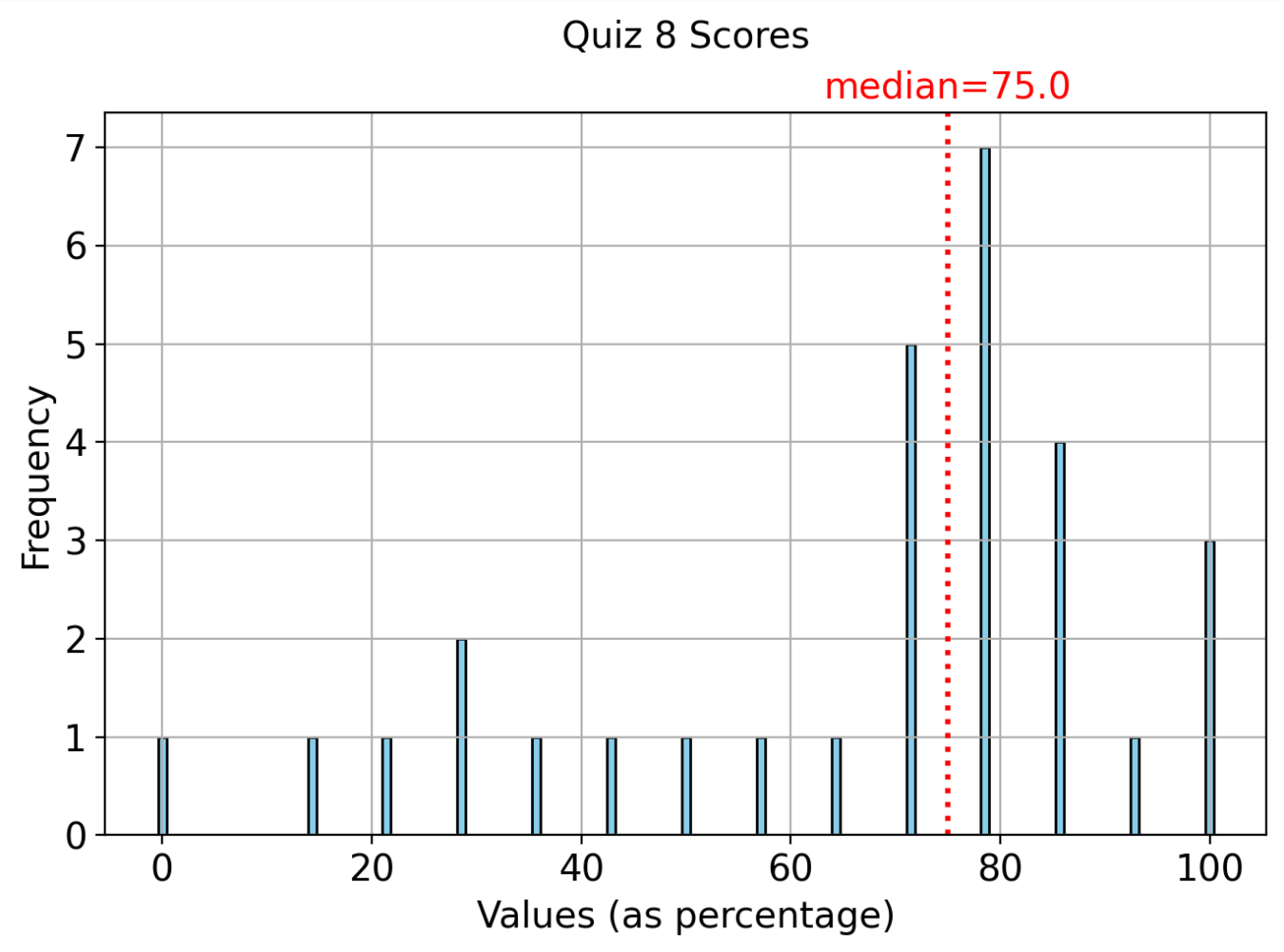
1. Fixed for loops to be Python (with exclusive for loops)
2. Why Big O is not worst case.
3. Why we like Big O/Big Theta.

Feedback on Weekly Quiz #8

Recall: Weekly Quiz #7



Weekly Quiz #8



Introduction to Probability: Samples and Events

Example: Drawing a card from a well-shuffled deck

Suit \ Rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
Hearts ♡	A♡	2♡	3♡	4♡	5♡	6♡	7♡	8♡	9♡	10♡	J♡	Q♡	K♡
Diamonds ♢	A♢	2♢	3♢	4♢	5♢	6♢	7♢	8♢	9♢	10♢	J♢	Q♢	K♢
Clubs ♣	A♣	2♣	3♣	4♣	5♣	6♣	7♣	8♣	9♣	10♣	J♣	Q♣	K♣

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Remark. A standard deck: 4 suits $\times$ 13 ranks $=$ 52 cards

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What is the probability of each outcome? Since each outcome is equally likely, each outcome has the same probability: $\frac{1}{52}$.

Finite probability spaces

Definition

A (finite) **probability space** is a pair (S, P) where S is a finite, nonempty set and P is a function $S \to \mathbb{R}$ such that $P(s) \geq 0$ for all $s \in S$ and

$$\sum_{s \in S} P(s) = 1$$

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Remark: What is a sample space?

We refer to S as the **sample space**: the set of all possible outcomes of a random process or experiment.

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⚠ Scheinerman refers to (S, P) as a *sample space*. However, most people use *sample space* to refer to S alone. (We will follow the latter convention.)

Example: Playing cards

Definition

An **event** is a subset of a sample space.

Suit \ Rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
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Poll. What is an example event when drawing a card from a well-shuffled deck?

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Poll. What is an example event when drawing a card from a well-shuffled deck?

Example. The event of drawing a 7 is given by $E_7 = \{7\spadesuit, 7\heartsuit, 7\diamondsuit, 7\clubsuit\}$.

Example: Playing cards

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An **event** is a subset of a sample space.

Suit \ Rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
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Poll. What is an example event when drawing a card from a well-shuffled deck?

Example. The event of drawing a 8 is given by $E_8 = \{8♠, 8♡, 8♢, 8♣\}$.

Example: Playing cards

Definition

An **event** is a subset of a sample space.

Suit \ Rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
Hearts ♡	A♡	2♡	3♡	4♡	5♡	6♡	7♡	8♡	9♡	10♡	J♡	Q♡	K♡
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Poll. What is an example event when drawing a card from a well-shuffled deck?

Example. The event of drawing a 7 or 8 is given by $E_{7 \text{ or } 8} = \{7♠, 7♡, 7♢, 7♣, 8♠, 8♡, 8♢, 8♣\}$.

Example: Playing cards

Definition

An **event** is a subset of a sample space.

Suit \ Rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
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Poll. What is an example event when drawing a card from a well-shuffled deck?

Example. The event of drawing a 7 or 8 is given by $E_{7 \text{ or } 8} = \{7♠, 7♡, 7♢, 7♣, 8♠, 8♡, 8♢, 8♣\}$. Is there a more compact way to write this?

Example: Playing cards

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An **event** is a subset of a sample space.

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Example. The event of drawing a 7 or 8 is given by $E_{7 \text{ or } 8} = \{7♠, 7♡, 7♢, 7♣, 8♠, 8♡, 8♢, 8♣\}$. Is there a more compact way to write this?You could write this as $E_{7 \text{ or } 8} = E_7 \cup E_8$.

Example: Playing cards

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Poll. What is an example event when drawing a card from a well-shuffled deck?

Example. The event of drawing a heart is given by $E_{\heartsuit} = \{A\heartsuit, 2\heartsuit, \ldots, K\heartsuit\}$.

Example: Playing cards

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Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
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Example. The event of drawing a diamond is given by $E_{\diamondsuit} = \{A\diamondsuit, 2\diamondsuit, \ldots, K\diamondsuit\}$.

Example: Playing cards

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Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
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Clubs ♣	A♣	2♣	3♣	4♣	5♣	6♣	7♣	8♣	9♣	10♣	J♣	Q♣	K♣

Poll. What is an example event when drawing a card from a well-shuffled deck?

Example. The event of drawing a red card (a heart or diamond) is given by $E_{\text{red}} = \{A\heartsuit, 2\heartsuit, \ldots, K\heartsuit, A\diamondsuit, 2\diamondsuit, \ldots, K\diamondsuit\}$.

Example: Playing cards

Definition

An **event** is a subset of a sample space.

Suit \ Rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
Spades ♠	A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
Hearts ♡	A♡	2♡	3♡	4♡	5♡	6♡	7♡	8♡	9♡	10♡	J♡	Q♡	K♡
Diamonds ♢	A♢	2♢	3♢	4♢	5♢	6♢	7♢	8♢	9♢	10♢	J♢	Q♢	K♢
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Events and their probabilities

Definition

Let (S, P) be a probability space. Then an **event** E is a subset of S (i.e. $E \subseteq S$). The probability of an event E, denoted $P(E)$ is

$$P(E) = \sum_{s \in E} P(s) \tag{1}$$

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Example. In the cards example, what is the probability of getting a 7?

Let $E_7 = \{7\spadesuit, 7\heartsuit, 7\diamondsuit, 7\clubsuit\}$. Then

$$\begin{aligned} P(E_7) &= P(7\spadesuit) + P(7\heartsuit) + P(7\diamondsuit) + P(7\clubsuit) \\ &= \frac{1}{52} + \frac{1}{52} + \frac{1}{52} + \frac{1}{52} \\ &= \frac{4}{52} = \frac{1}{13}. \end{aligned}$$

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Example. In the cards example, what is the probability of getting a 7 or 8?

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Example. In the cards example, what is the probability of getting a 7 or 8? Let $E = \{7\spadesuit, 7\heartsuit, 7\diamondsuit, 7\clubsuit, 8\spadesuit, 8\heartsuit, 8\diamondsuit, 8\clubsuit\}$. Then

$$\begin{aligned} P(E) &= P(7\spadesuit) + P(7\heartsuit) + P(7\diamondsuit) + P(7\clubsuit) + P(8\spadesuit) + P(8\heartsuit) + P(8\diamondsuit) + P(8\clubsuit) \\ &= \frac{1}{52} + \frac{1}{52} + \frac{1}{52} + \frac{1}{52} + \frac{1}{52} + \frac{1}{52} + \frac{1}{52} + \frac{1}{52} \\ &= \frac{8}{52} = \frac{2}{13}. \end{aligned}$$

A shortcut that you can use (sometimes!)

Proposition: Equally Likely Probability Formula

If (P, S) is a probability space in which all outcomes are equally likely and E is an event in S, then the probability of E is

$$P(E) = \frac{|E|}{|S|} = \frac{\text{the number of outcomes in } E}{\text{the number of outcomes in } S}$$

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$$\begin{aligned} P(E) &= \frac{|E|}{|S|} \\ &= \frac{|\{7\spadesuit, 7\heartsuit, 7\diamondsuit, 7\clubsuit, 8\spadesuit, 8\heartsuit, 8\diamondsuit, 8\clubsuit\}|}{|\texttt{whole deck}|} \\ &= \frac{8}{52} = \frac{2}{13}. \end{aligned}$$

Poll (Practice Reading Quiz)

A *hand* of poker is a five-element subset of the standard deck of 52 cards. Assuming that the deck is well-shuffled, what is the probability of drawing the hand $6\clubsuit - K\heartsuit - 3\diamondsuit - A\spadesuit - 7\clubsuit$?

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Solution

Since all poker hands are equally likely, we can use the equally likely probability formula. Let S be the set of all possible poker hands, and E be the (one-element) set containing the specific poker hand above.

Then

$$P(E) = \frac{|E|}{|S|} = \frac{\text{the number of outcomes in } E}{\text{the number of outcomes in } S} = \frac{1}{\binom{52}{5}} = \frac{1}{2,598,960}.$$

Poll

A bag contains 20 marbles. These marbles are identical, except that they are labeled with the integers 1 through 20. Five marbles are drawn randomly from the bag. What is the probability space?



Solution(s): The solution depends on what's meant by "randomly"!

There are a few ways to think about this.

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3. *The marbles are drawn one at a time with replacement.* Here, we again consider the outcomes as lists, but now $(1,1,1,1,1)$ is possible.

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$$S = \{1,\ldots,20\}^5, |S| = 20^5, P(s) = \frac{1}{20^5} \; \forall s \in S$$

Group exercises

Random group assignments

Ashley, Isabelle: 2
Bailey, Tanna: 6
Blaisdell, Olyver: 8
Brown, Ashton: 2
Cardenas, Dominic: 3
Cutler, George: 1
Fabik, Roland: 1
Fellows, Audrey: 10
Fitzgerald, Ryan: 7
Fuller, Ryan: 3
Griffen, Clara: 2
Grutsch, Tanner: 11
Hager, Nathan: 9
Horner, Jayden: 12
Johnson, Jaret: 1
Kintzel, Samantha: 6
Lameres, Alexis: 7
Lee, Holden: 11
Mayala, Reece: 9
McLean, Connor: 6
Mcbride, Daniel: 11
Miller, Maverick: 5
Moss, Emily: 12
Mousseau, Matthew: 10
Neuman, Evan: 12
Putnam, John: 5
Ramos, Vydel: 3
Rathbun, Jedidiah: 7
Roberts, Matthew: 10
Roller, Nathan: 8
Ryan, Otis: 5
Shevtsov, Timothy: 4
Stensland, Emma: 4
Thiessen, Logan: 8
Thompson, Jaiden: 4
Vath, Ethan: 9

Group exercises

1. Let (S, P) be the probability space in which $S = \{1, 2, 3, 4\}$. Suppose $P(1) = x$, $P(2) = 2x$, $P(3) = 3x$, and $P(4) = 4x$. Find x.

2. Suppose that that (S, P_1) and (S, P_2) are two probability spaces that have the same set of outcomes, S. Is it possible that each outcome is less likely in the first probability space than it is in the second? That is, can we have $\forall s \in S, P_1(s) < P_2(s)$?

3. A dart is thrown blindly at the target shown in the figure. The probability that the dart lands in one of the four concentric regions is proportional to the area of the region. The radii of the circles in the figure are 1,2,3, and 4 units, respectively. Please note that region 2 consists of just the annular region from radius 1 to 2, and does not include the enclosed circular region 1. Let (S, P) be a probability space modeling this situation. The set S consists of the four outcomes: hitting region 1,2,3 or 4. Please find $P(1), P(2), P(3)$, and $P(4)$.



Solution to group exercise #1

Problem. Let (S, P) be the probability space in which $S = \{1, 2, 3, 4\}$. Suppose $P(1) = x, P(2) = 2x, P(3) = 3x$, and $P(4) = 4x$. Find x.

Solution.

$$\sum_{s \in S} P(s) = 1 \qquad \text{(Def. probability space)}$$

$$\implies P(1) + P(2) + P(3) + P(4) = 1 \qquad \text{(Expand sum)}$$

$$\implies x + 2x + 3x + 4x = 1 \qquad \text{(By assumption)}$$

$$\implies 10x = 1$$

$$\implies x = \frac{1}{10}$$

Remark.

Therefore,

$$P(1) = \frac{1}{10}, \quad P(2) = \frac{2}{10}, \quad P(3) = \frac{3}{10}, \quad P(4) = \frac{4}{10}.$$

Solution to group exercise #2

Problem. Suppose that that (S, P_1) and (S, P_2) are two probability spaces that have the same set of outcomes, S. Is it possible that each outcome is less likely in the first probability space than it is in the second? That is, can we have $\forall s \in S, P_1(s) < P_2(s)$?

Solution. No. By definition of a (finite) probability space, we have

$$\sum_{s \in S} P_1(s) = \sum_{s \in S} P_2(s) = 1 \tag{2}$$

But since $P_1(s) < P_2(s)$ for all s, we have

$$\sum_{s \in S} P_1(s) < \sum_{s \in S} P_2(s) \tag{3}$$

Substituting Eq. (2) into Eq. (3) , we obtain

$$1 < 1$$

which is a contradiction.

Remark on group exercise #2

Proof Template 12 Proof by contradiction

To prove "If A, then B":
We assume the conditions in A.
Suppose, for the sake of contradiction, not B.
Argue until we reach a contradiction.
$\Rightarrow\Leftarrow$ ■

(The symbol $\Rightarrow\Leftarrow$ is an abbreviation for the following: Thus we have reached a contradiction. Therefore the supposition (not B) must be false. Hence B is true.)

Figure 1: *A template for proof by contradiction.* Source: Scheinerman Sec. 20 (pp. 121)

Remark

This is an example of proof by contradiction (Scheinerman Sec. 20).

Proposition A:	(S, P_1) and (S, P_2) are two probability spaces that have the same set of outcomes, S.
Proposition B:	$\forall s \in S, P_1(s) < P_2(s)$

We prove $A \implies$ (`not` B) by assuming A and B and obtaining a contradiction.

Solution to group exercise #3

The areas of the circles are given by

$$C_1 = \pi 1^2 = \pi \qquad C_3 = \pi 3^2 = 9\pi$$
$$C_2 = \pi 2^2 = 4\pi \qquad C_4 = \pi 4^2 = 16\pi$$

Hence, the areas of the annular regions are given by

$$A_1 = C_1 = \pi \qquad A_3 = C_3 - C_2 = 5\pi$$
$$A_2 = C_2 - C_1 = 3\pi \qquad A_4 = C_4 - C_3 = 7\pi$$

By assumption, the probability of the dart landing in each annular region is

$$P(1) \propto A_1, \quad P(2) \propto A_2, \quad P(3) \propto A_3, \quad P(4) \propto A_4,$$

where $\propto$ means "is proportional to". In other words,

$$P(1) = \frac{A_1}{Z}, \quad P(2) = \frac{A_2}{Z}, \quad P(3) = \frac{A_3}{Z}, \quad P(4) = \frac{A_4}{Z},$$

where Z is some unknown "normalizing constant". We solve for Z by using that the probabilities over all outcomes must sum to unity:

$$\sum_{s \in S} P(s) = 1 \implies \frac{A_1}{Z} + \frac{A_2}{Z} + \frac{A_3}{Z} + \frac{A_4}{Z} = 1$$

$$\implies A_1 + A_2 + A_3 + A_4 = Z \implies Z = 16\pi.$$

Hence

$$P(1) = \frac{1}{16}, \quad P(2) = \frac{3}{16}, \quad P(3) = \frac{5}{16}, \quad P(4) = \frac{7}{16}.$$